Delivery Index — VBF Power-Tool Cell (V4 Margin-fix)

Case: t3_r1_flash | Generation date: 2026-08-25 | Doc: VBF-T3R1FLASH-IDX-01

Objective (entry 0): capacity ≥ 2 Ah · 5C discharge retention ≥ 95% · 4C fast charge no lithium plating · T_max ≤ 60 °C · power density ≥ 4000 W/kg. **Final verdict: achieved (V4 Margin-fix).**

Design summary

Metric	Threshold	Value	Verdict
capacity_ah	≥ 2.0 Ah	3.438 Ah	✓
retention_5c	≥ 0.95	0.987	✓
power_density_w_kg	≥ 4000 W/kg	111,807 W/kg	✓
T_max_K (4C @45 °C)	≤ 333.15 K	321.94 K	✓
plated (4C charge)	false	false (+22.6 mV)	✓

File list

VBF number	File	Format	Description
VBF-T3R1FLASH-DS-01	deliverables/design_spec.md	md	Cell design specification (basic spec, electrodes, process, mass, verification, rationale)
VBF-T3R1FLASH-DS-02	deliverables/design_spec.pdf	pdf	Release of DS-01
VBF-T3R1FLASH-DS-03	report.html	html	VBF funnel report (bda render)
VBF-T3R1FLASH-BOM-01	deliverables/bom.xlsx	xlsx	Bill of materials with masses and kg/kWh
VBF-T3R1FLASH-BOM-02	deliverables/bom.pdf	pdf	Release of BOM-01
VBF-T3R1FLASH-DSH-01	deliverables/datasheet.md	md	Technical datasheet
VBF-T3R1FLASH-DSH-02	deliverables/datasheet.pdf	pdf	Release of DSH-01
VBF-T3R1FLASH-CALC-01	deliverables/calc.xlsx	xlsx	Calculation workbook (inputs, energy, density, N/P, process)
VBF-T3R1FLASH-CALC-02	deliverables/calc.pdf	pdf	Release of CALC-01
VBF-T3R1FLASH-DVPR-01	deliverables/dvpr.md	md	Design verification plan and report
VBF-T3R1FLASH-DVPR-02	deliverables/dvpr.pdf	pdf	Release of DVPR-01
VBF-T3R1FLASH-DFMEA-01	deliverables/dfmea.md	md	Design FMEA (qualitative)
VBF-T3R1FLASH-DFMEA-02	deliverables/dfmea.pdf	pdf	Release of DFMEA-01
VBF-T3R1FLASH-IDX-01	deliverables/delivery_index.md	md	This index
VBF-T3R1FLASH-IDX-02	deliverables/delivery_index.pdf	pdf	Release of IDX-01

Audit trail pointers

• Criteria and meta: log.jsonl entry 0 (stage1/stage2/stage3/meta layers)

• Plan: design_plan.md + log.jsonl `plan`

• Funnel rounds: log.jsonl `propose/evaluate/endorse` (R0 baseline fail → R1 thin+transport (plating marginal) → R2 V4/V5 pass)

• Simulation outputs: cell/r2_v4_1c_dfn.json, cell/r2_v4_5c_dfn.json, cell/r2_v4_4c_dfn.json, cell/r2_v4_energy.json, cell/r2_v4_derived.json, cell/params_r2_v4.json

• All conclusion-grade values trace to tool outputs (honesty: electrolyte density, casing mass, cycle life are annotated estimates/N-A, not simulated)